

```
1 import pandas as pd
2 from sklearn.cluster import KMeans
3 import matplotlib.pyplot as plt
4 data = {
5     "Customer_ID": [101,102,103,104,105,106,107,108],
6     "Amount_Spent": [200, 800, 150, 900, 400, 700, 300, 1000],
7     "Items_Purchased": [2, 8, 1, 9, 4, 7, 3, 10]
8 }
9 df = pd.DataFrame(data)
10 X = df[["Amount_Spent", "Items_Purchased"]]
11 kmeans = KMeans(n_clusters=3, random_state=0)
12 df["Cluster"] = kmeans.fit_predict(X)
13 plt.scatter(df["Amount_Spent"], df["Items_Purchased"], c=df["Cluster"], cmap='viridis')
14 plt.xlabel("Amount Spent")
15 plt.ylabel("Items Purchased")
16 plt.title("Customer Segmentation using K-Means")
17 plt.show()
18 print(df)
```

	Customer_ID	Amount_Spent	Items_Purchased	Cluster
0	101	200	2	0
1	102	800	8	2
2	103	150	1	0
3	104	900	9	1
4	105	400	4	0
5	106	700	7	2
6	107	300	3	0
7	108	1000	10	1

Figure 1

